



CL-1002

Programming Fundamentals

Lab # 2

Objectives:

- Introduction to procedural flow.
- Exhibit the understanding of pseudocode.
- Exhibit the understanding of drawing Flow Charts.

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your logic on Copy / Notebook.
2. Write pseudocode in handwritten on **Paper using Pen**.
3. Write **Your Name** and **Roll No** on your Paper/Sheet's first page.
4. **Do not copy from any source otherwise you will be penalized with negative marks.**
5. Complete your lab **within given Time Slot**.

Problem: Write pseudocode of decision-based problems.

1.1 Sequences Problems

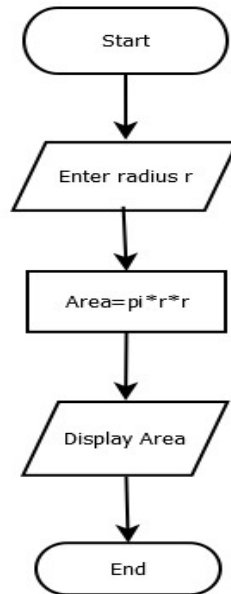
1. Write a program logic that calculate the total bill of the shopping, a person purchases two keyboards each worth of 100\$, three mouse each worth of 50\$. Calculate the Total in PKR.
2. A brand offered 10% discount on each shirt purchased. The original price of the shirt is 550, find the discounted price of one shirt. Write the pseudocode for this problem statement.

1.2 Decision Problems

1. Write pseudocode to print positive number entered by the user. If user enters a negative number, it is skipped.
2. Write pseudocode to add bonus of 100\$ to each person whose salary is greater than 1500\$.
3. Write pseudocode to check whether a character is Vowel or Consonant.
4. Write pseudocode to that year input from user and check whether it's Leap Year or Not.
5. Write a pseudocode of a program that finds the greatest of three number.

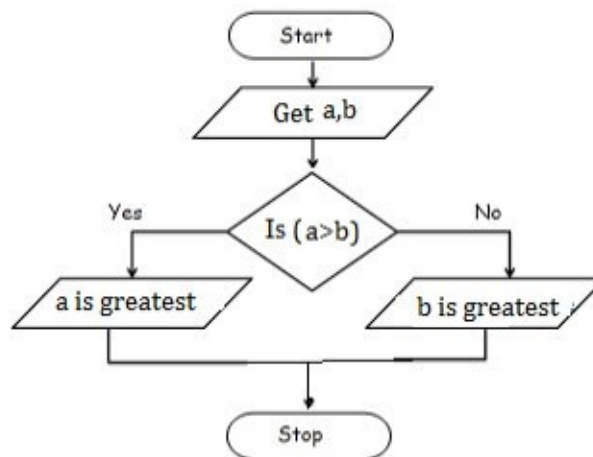
2. FLOWCHARTS

Problem-1: Calculate the area of a circle.



Program-2: Draw flow chart of a program that takes two numbers as input and outputs as “divisible” if number-1 is divisible by number-2 otherwise outputs “not divisible”.

Problem-3: Draw the flow chart for finding the greatest of two numbers



1. Draw flow chart to check of an entered number of even or odd.
2. Draw flow chart of a program that finds the greatest of three number.
3. Draw flow chart of a program that checks exams of two students. Program checks of two exams have more than 60% similarity then both students are declared as fail. Otherwise checks of a student has more than 50% marks then student is declared as pass, if student has less than 50% marks then student is declared as fail.
4. Draw a flow chart for a program that asks the user for choice of shape. If user enters 'circle' as choice then it asks you to enter 'radius' and program return area of the circle. If user enters 'triangle' as choice then program asks you enter the length of sides and returns the parameter of the triangle. If user enters 'rectangle' then program asks you to enter height and width and finally returns the area of the rectangle.

Best of Luck 😊

You need to done with your exercise within given time.